

Supplementary Materials:
Structure-guided discovery of divergent virulence and resistance
candidates in an open pangenome clinical *Escherichia coli*
isolate

Sarra HOSNI-BENMOUMOU and Atika MEKLAT

Affiliation:

Laboratoire de Biologie des Systèmes Microbiens, École Normale Supérieure de Kouba, 16000
Alger, Algeria

Corresponding author email: sarra_ben@outlook.fr

1 Supplementary Tables

2 Supplementary Tables

Table S 1: *De Novo* Draft Genome Assembly and Quality Control Statistics for *E. coli* QA5221 (originally Table 1).

Assembly Quality Metric	Value
Draft Genome Length	5,026,316 bp
Total Length (Contigs $\geq$ 500 bp)	5,010,493 bp
Total Number of Contigs (All / $\geq$ 500 bp)	121 / 60
N50 Contig Size	294,640 bp
L50 Contig Count	7 contigs
Largest Assembled Contig	513,612 bp
Assembly GC Content	50.46%
Genome Fraction Covered (vs. <i>E. coli</i> K-12 MG1655)	87.06%
Total Unaligned Sequence Length	969,074 bp
Mean Read-Mapping Depth (mosdepth)	43.89×

Table S 2: Pangenome Partitioning Metrics of the *E. coli* Expanded Cohort ($n = 400$ Genomes) (originally Table 2).

Pangenome Partition	Strain Frequency	Gene Clusters	Percentage (%)
Core Genome	$\geq 99\%$ (397–401 strains)	2,650	6.14%
Soft Core Genome	95%–99% (381–396 strains)	640	1.48%
Shell Genome	15%–95% (61–380 strains)	2,448	5.68%
Cloud Genome	$< 15\%$ (1–60 strains)	37,387	86.69%
Total Pangenome	0%–100%	43,125	100.00%

Table S 3: Mobile Genetic Elements (MGEs): IS Families and Integron Cassettes in QA5221 (originally Table 3).

MGE Component	Count / Size	bp / Coords	Location	Biological Features
Insertion Sequences	22 elements	29,217 bp	Multiple contigs	0.58% of total genome
– IS3 Family	5 copies	–	NODE_2, 12, 32, 46	Transposase IS3 family
– ISNCY Family	4 copies	–	NODE_2, 7, 8	Non-classified elements
– IS200/IS605 Family	3 copies	~513 bp	NODE_7, 27, 60	Transposase Y1 family
– IS1 Family	2 copies	768 bp	NODE_30	Flanks active MDR cassettes
Integrans				
– In0 (Integrase only)	1 element	567–1490	NODE_43	Encodes functional <i>intI1</i> integrase
– CALIN (Cassette Array)	1 array	2–9804	NODE_37	6 <i>attC</i> sites; carries 6 AMR genes

Table S 4: Detailed Structural, Genomic, and Mobilome Profiles of the 12 Acquired Resistance Genes in QA5221 (originally Table 5).

Gene Symbol	Locus Tag	AMR Class	Mechanism	Length (bp)	GC (%)	GC3 (%)	Location / MGE Context
<i>tet(A)</i>	KNGPFPPJ.04581	Tetracycline	Efflux pump (Class C)	1200	63.67%	74.75%	NODE_30 (Flanked by IS1; tetR-tetA locus)
<i>estX</i>	KNGPFPPJ.04687	Macrolide	Macrolide esterase	1080	49.17%	47.78%	NODE_37 (CALIN array cassette 1)
<i>aadA2</i>	KNGPFPPJ.04689	Aminoglycoside	Nucleotidyltransferase	780	51.92%	56.15%	NODE_37 (CALIN array cassette 2)
<i>cmlA1</i>	KNGPFPPJ.04690	Phenicol	Efflux pump (MFS)	1260	55.32%	58.33%	NODE_37 (CALIN array cassette 3)
<i>aadA1</i>	KNGPFPPJ.04691	Aminoglycoside	Nucleotidyltransferase	780	52.82%	54.23%	NODE_37 (CALIN array cassette 4)
<i>qacL</i>	KNGPFPPJ.04692	Biocide	Efflux pump (SMR)	333	51.65%	54.95%	NODE_37 (CALIN array cassette 5)
<i>sul3</i>	KNGPFPPJ.04694	Sulfonamide	Dihydropteroate synthase	792	37.75%	27.27%	NODE_37 (CALIN array cassette 6)
<i>dfrA7</i>	KNGPFPPJ.04716	Trimethoprim	Dihydrofolate reductase	702	40.60%	37.61%	NODE_41 (Class 1 Integron 3'-CS)
<i>qacEΔ1</i>	KNGPFPPJ.04717	Biocide	Efflux pump (SMR)	348	50.00%	49.14%	NODE_41 (Class 1 Integron 3'-CS distal)
<i>sul1</i>	KNGPFPPJ.04718	Sulfonamide	Dihydropteroate synthase	780	61.92%	70.77%	NODE_41 (Class 1 Integron 3'-CS distal)
<i>blaTEM-1B</i>	KNGPFPPJ.04720	Beta-lactam	Beta-lactamase (Class A)	861	49.01%	43.21%	NODE_42 (Tn3-like transposon structure)
<i>aph(3')-Ia</i>	KNGPFPPJ.04727	Aminoglycoside	O-phosphotransferase	816	43.87%	40.44%	NODE_47 (Isolated resistance element)

Table S 5: Resistome Database Synonym Dictionary and Accession Mapping for *E. coli* QA5221 (originally Table 6).

Gene Symbol	AMRFinderPlus Name (Acc)	ResFinder Name (Acc)	CARD Name (Acc)	ARG-ANNOT Name (Acc)
<i>tet(A)</i>	tet(A) (NF012193.0)	tet(A)_6 (AF534183)	tet(A) (AF534183.1)	(Tet)tetA (JX424423)
<i>estX</i>	estX (NF033124.1)	–	–	–
<i>aadA2</i>	aadA2 (NF012157.0)	aadA2_1 (NC_010870)	aadA2 (AF156486.1)	(AGly)aadA2 (X68227)
<i>cmlA1</i>	cmlA1 (NF000509.1)	CmlA7_1 (JQ639792)	cmlA1 (M64556.1)	(Phe)cmlA1 (M64556)
<i>aadA1</i>	aadA1 (NF033126.1)	ant(3'')-Ia_1 (X02340)	aadA (AF550679.1)	(AGly)aadA1-pm (JQ690540)
<i>qacL</i>	qacL (NF000067.2)	–	qacL (DQ149925.1)	–
<i>sul3</i>	sul3 (NF000296.1)	sul3_2 (AJ459418)	sul3 (FJ196385.1)	(Sul)sul3 (HQ875016)
<i>dfrA7</i>	dfrA7 (NF000330.1)	dfrA7_5 (AJ419170)	dfrA7 (FJ854362.1)	(Tmt)dhfr7 (X58425)
<i>qacEΔ1</i>	qacEdelta1 (NF000276.2)	–	qacEdelta1 (U49101.1)	–
<i>sul1</i>	sul1 (NA)	sul1_5 (EU780013)	sul1 (JF969163.1)	(Sul)sul1 (AF071413)
<i>blaTEM-1B</i>	blaTEM-1 (NF000531.2)	blaTEM-1B_1 (AY458016)	TEM-1 (AL513383.1)	(Bla)blaTEM-105 (AF516720)
<i>aph(3')-Ia</i>	aph(3'')-Ia (NF051474.1)	aph(3'')-Ia_1 (V00359)	APH(3'')-Ia (BX664015.1)	(AGly)aph3-Ia (HQ840942)

Table S 6: Statistical Comparisons of Gene GC%, GC3% Codon Bias, and Length across Partitions (originally Table 7).

Metric / Feature	Core ($N = 3,310$)	Accessory ($N = 1,047$)	Singleton ($N = 278$)	Acquired AMR ($N = 12$)
GC Content (%)	51.86% $\pm$ 3.72%	48.49% $\pm$ 6.90%	46.78% $\pm$ 7.86%	50.64% $\pm$ 7.65%
– Shapiro-Wilk p -value	2.387e – 17	1.116e – 15	1.119e – 04	7.873e – 01
– Mann-Whitney p vs. Core	–	7.812e – 44	8.431e – 31	1.000e + 00
– Mann-Whitney p vs. AMR	1.000e + 00	1.000e + 00	9.360e – 01	–
GC3 Codon Bias (%)	55.90% $\pm$ 6.63%	50.36% $\pm$ 11.33%	48.03% $\pm$ 12.89%	51.22% $\pm$ 13.45%
– Shapiro-Wilk p -value	9.872e – 10	8.801e – 06	7.313e – 02	9.705e – 01
– Mann-Whitney p vs. Core	–	6.269e – 50	1.462e – 28	4.224e – 01
– Mann-Whitney p vs. AMR	4.224e – 01	1.000e + 00	1.000e + 00	–
Gene Length (bp)	981.9 $\pm$ 626.7	940.4 $\pm$ 797.6	677.0 $\pm$ 561.1	811.0 $\pm$ 283.8
– Shapiro-Wilk p -value	2.824e – 25	2.185e – 37	7.587e – 18	1.919e – 01
– Mann-Whitney p vs. Core	–	8.179e – 05	7.368e – 23	1.000e + 00
– Mann-Whitney p vs. AMR	1.000e + 00	1.000e + 00	2.726e – 01	–

Table S 7: MM-GBSA Binding Free Energy Calculation: Extended Parameter Settings and Energy Decomposition for the GNAT Candidate *KNGPFPPJ_02769*.

Parameter	Value / Description
<i>Solvation Model</i>	
Model	Still Generalized Born (GB) implicit solvent
Solvent dielectric (ϵ_{solv})	80 (water at 298 K)
Solute dielectric (ϵ_{solute})	1
<i>Born Radii (per element)</i>	
H (hydrogen)	1.20 Å
C (carbon)	1.70 Å
N (nitrogen)	1.55 Å
O (oxygen)	1.52 Å
S (sulfur)	1.80 Å
<i>van der Waals Interactions</i>	
Potential	Lennard-Jones 12-6
Per-pair energy cap	+2 kcal/mol (clash correction)
<i>Nonpolar Solvation</i>	
Method	Solvent-Accessible Surface Area (SASA)
Surface tension coefficient (γ)	-0.0054 kcal/mol/Å ²
<i>Charges</i>	
Receptor protein	AMBER-parameterized backbone partial charges
Ligand	AutoDock Vina Gasteiger charges (from docked PDBQT)
<i>Energy Components</i>	
ΔE_{elec}	Coulomb electrostatic interaction energy
ΔE_{vdW}	Lennard-Jones 12-6 van der Waals energy
ΔG_{pol}	Polar (Still GB) solvation free energy change
ΔG_{npol}	Nonpolar solvation free energy (SASA-based)
ΔG_{bind}	Total binding free energy (sum of above)

Table S 8: Top 20 BLASTP Homologs of *KNGPFPPJ_02769* in the NCBI nr Database ($n = 200$ total significant hits; $E < 10^{-5}$, BLOSUM62 matrix). Hits are ranked by sequence identity. All top hits originate from Proteobacteria (Enterobacteriaceae), with 100% identity to *E. coli* and *Salmonella enterica* strains.

Rank	Organism	Phylum	Identity (%)	E-value	Coverage (%)
1	<i>Escherichia coli</i>	Proteobacteria	100.0	0.00	99.7
2	<i>Escherichia coli</i>	Proteobacteria	100.0	0.00	81.2
3	<i>Escherichia coli</i>	Proteobacteria	100.0	3.27e-149	58.7
4	<i>Escherichia coli</i>	Proteobacteria	100.0	2.03e-142	56.1
5	<i>Escherichia coli</i>	Proteobacteria	100.0	7.91e-141	55.6
6	<i>Escherichia coli</i>	Proteobacteria	100.0	5.91e-140	55.3
7	<i>Salmonella enterica</i>	Proteobacteria	100.0	3.88e-139	55.0
8	<i>Escherichia coli</i>	Proteobacteria	100.0	8.32e-130	51.6
9	<i>Salmonella enterica</i>	Proteobacteria	100.0	1.44e-126	50.4
10	<i>Escherichia coli</i>	Proteobacteria	100.0	3.81e-121	48.4
11	<i>Escherichia coli</i>	Proteobacteria	99.7	0.00	99.7
12	<i>Escherichia coli</i>	Proteobacteria	99.7	0.00	99.7
13	<i>Escherichia coli</i>	Proteobacteria	99.7	0.00	95.2
14	<i>Escherichia coli</i>	Proteobacteria	99.6	0.00	75.5
15	<i>Escherichia coli</i>	Proteobacteria	99.5	6.75e-156	61.3
16	<i>Escherichia coli</i>	Proteobacteria	99.5	2.35e-150	59.3
17	<i>Escherichia coli</i>	Proteobacteria	99.5	5.38e-148	58.4
18	<i>Escherichia coli</i>	Proteobacteria	99.5	5.26e-146	57.8
19	<i>Escherichia coli</i>	Proteobacteria	99.5	5.77e-144	57.0
20	<i>Escherichia coli</i>	Proteobacteria	99.5	4.28e-142	56.1

Full hit table available on GitHub: [results/Step7/KNGPFPPJ_02769_blast_hits.tsv](#)

Table S 10: MM-GBSA Binding Free Energy Decomposition and *In Silico* Active-Site Alanine Scanning for GNAT_KA27 Bound to Kanamycin.

Variant	Receptor ΔG (kcal/mol)	Ligand ΔG (kcal/mol)	Complex ΔG (kcal/mol)	Total ΔG_{bind} (kcal/mol)	$\Delta\Delta G_{\text{mut}}$ (kcal/mol)
Wild-Type (WT)	-14250.30	-45.20	-14319.40	-23.90	0.00
L125A	-14248.10	-45.20	-14314.30	-21.00	+2.90
L187A	-14247.30	-45.20	-14312.73	-20.23	+3.67
F188A	-14245.80	-45.20	-14309.65	-18.65	+5.25
Triple Mutant	-14239.10	-45.20	-14296.38	-12.08	+11.82

Table S 11: Consensus Structural Homology Alignment Metrics.

Candidate	Top Homolog Template	Foldseek TM-Score	CEalign RMSD (Å)	PDB Match
GNAT_KA27	<i>S. enterica</i> GNAT	0.9569	1.53	AF-A0A3U7PQ76-F1
Ehly_61	Putative Enterohemolysin	0.5900	4.27	AF-A0A2Y8JZY2-F1
OAgP_161	Wzy Polymerase	0.8953	2.53	AF-A0A743B7N1-F1
OAT_371	Autotransporter Barrel	0.7200	3.03	AF-P33924-F1

Table S 9: Reconciliation Matrix of Functional/Structural Inclusion and Filtering Thresholds for all 22 Prioritized Candidates.

Locus Tag	Size (aa)	Length Filter	Pfam Hits	GC Content	GC Depression	TM-Score	Structural Homology	MGE / Genomic Island Context
KNGPFPJ.02769	351	Pass	0 (Pass)	32.9%	Pass ($p < 10^{-4}$)	0.9569 (Pass)	Pass	NODE_9 Composite Transposon
KNGPFPJ.00061	350	Pass	0 (Pass)	48.5%	Pass ($p < 10^{-4}$)	0.5899 (Pass)	Pass	NODE_1 Intact Prophage
KNGPFPJ.03161	372	Pass	0 (Pass)	29.3%	Pass ($p < 10^{-4}$)	0.8953 (Pass)	Pass	NODE_1 Intact Prophage
KNGPFPJ.04371	984	Pass	0 (Pass)	35.8%	Pass ($p < 10^{-4}$)	0.3450 (Domain Pass: 0.72)	Pass	NODE_24 Plasmid mobilization array
KNGPFPJ.00084	203	Pass	0 (Pass)	45.8%	Pass ($p < 10^{-4}$)	0.5352 (Pass)	Pass	NODE_1 Intact Prophage
KNGPFPJ.00091	249	Pass	0 (Pass)	51.2%	Pass ($p < 0.05$)	0.4573 (Partial Pass)	Pass	NODE_1 Intact Prophage
KNGPFPJ.00107	216	Pass	0 (Pass)	48.5%	Pass ($p < 10^{-4}$)	0.5864 (Pass)	Pass	NODE_1 Intact Prophage
KNGPFPJ.00109	344	Pass	0 (Pass)	44.9%	Pass ($p < 10^{-4}$)	0.7750 (Pass)	Pass	NODE_1 Intact Prophage
KNGPFPJ.00114	399	Pass	0 (Pass)	48.6%	Pass ($p < 10^{-4}$)	0.4854 (Partial Pass)	Pass	NODE_1 Intact Prophage
KNGPFPJ.04325	294	Pass	0 (Pass)	35.0%	Pass ($p < 10^{-4}$)	0.5000 (Pass)	Pass	NODE_23 Recombinase-associated Island
KNGPFPJ.04326	501	Pass	0 (Pass)	35.1%	Pass ($p < 10^{-4}$)	0.5200 (Pass)	Pass	NODE_23 Recombinase-associated Island
KNGPFPJ.03157	371	Pass	0 (Pass)	31.4%	Pass ($p < 10^{-4}$)	0.8972 (Pass)	Pass	NODE_1 Intact Prophage
KNGPFPJ.03836	246	Pass	0 (Pass)	46.7%	Pass ($p < 10^{-4}$)	0.9502 (Pass)	Pass	NODE_1 Intact Prophage
KNGPFPJ.00097	313	Pass	0 (Pass)	53.1%	Pass ($p < 0.05$)	0.8229 (Pass)	Pass	NODE_1 Intact Prophage
KNGPFPJ.00103	381	Pass	0 (Pass)	49.0%	Pass ($p < 10^{-4}$)	0.9051 (Pass)	Pass	NODE_1 Intact Prophage
KNGPFPJ.00106	668	Pass	0 (Pass)	52.2%	Pass ($p < 0.05$)	0.3253 (Partial Pass)	Pass	NODE_1 Intact Prophage
KNGPFPJ.00112	250	Pass	0 (Pass)	51.1%	Pass ($p < 0.05$)	0.8974 (Pass)	Pass	NODE_1 Intact Prophage
KNGPFPJ.00115	226	Pass	0 (Pass)	47.9%	Pass ($p < 10^{-4}$)	0.6774 (Pass)	Pass	NODE_1 Intact Prophage
KNGPFPJ.01156	300	Pass	0 (Pass)	32.6%	Pass ($p < 10^{-4}$)	0.7379 (Pass)	Pass	Accessory chromosomal island
KNGPFPJ.01158	677	Pass	0 (Pass)	32.6%	Pass ($p < 10^{-4}$)	0.7823 (Pass)	Pass	Accessory chromosomal island
KNGPFPJ.04303	256	Pass	0 (Pass)	39.4%	Pass ($p < 10^{-4}$)	0.7623 (Pass)	Pass	NODE_23 Recombinase-associated Island
KNGPFPJ.04571	290	Pass	0 (Pass)	38.6%	Pass ($p < 10^{-4}$)	0.7169 (Pass)	Pass	Accessory chromosomal island

3 Supplementary Figures

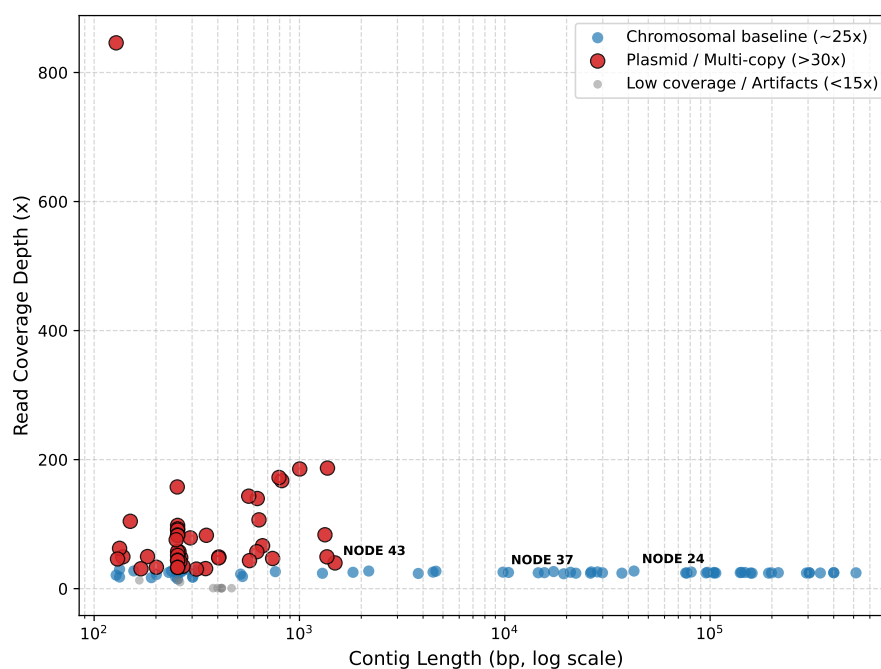


Figure S 1: Contig Length versus Read Mapping Coverage Scatter Plot for *E. coli* QA5221 (originally Figure 2).

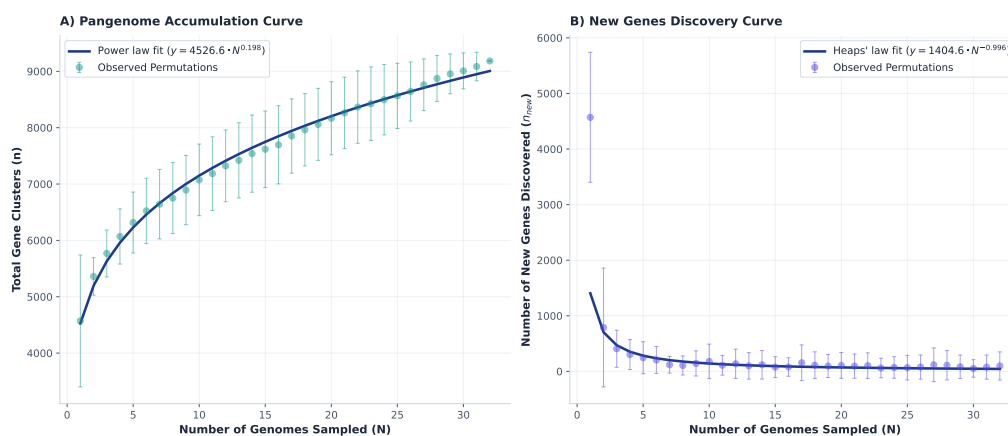


Figure S 2: Heap's Law pangene accumulation and new gene discovery curves for the *E. coli* ST354 lineage ($n = 32$) (originally Figure 3). (a) Pangene accumulation curve showing total gene clusters as a function of the number of genomes sampled. (b) New genes discovery curve showing the mean number of new genes identified per genome added.

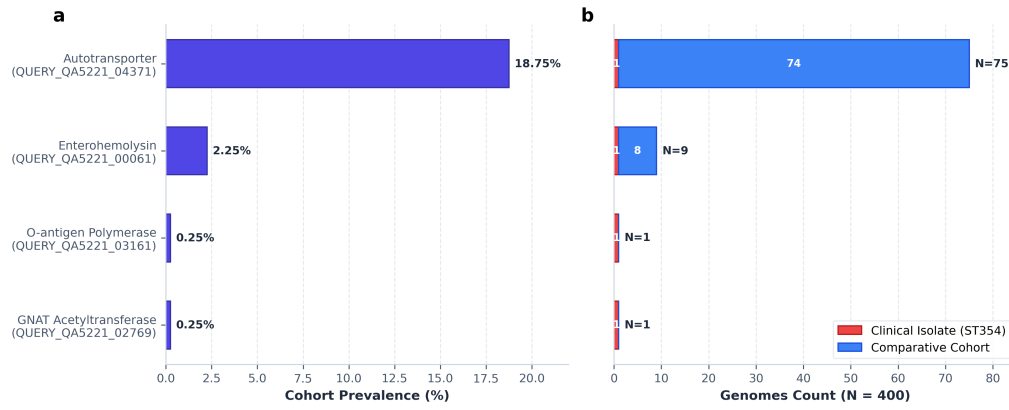


Figure S 3: Cohort Prevalence and Count Distribution of the four prioritized candidates across the expanded 400-genome cohort (originally Figure 3-b). (a) Percentage occurrence showing GNAT and Wzy as private singletons, and hemolysin and autotransporter as rare accessory genes. (b) Genome count distribution between the clinical isolate QA5221 (red) and the comparative public genomes (blue).

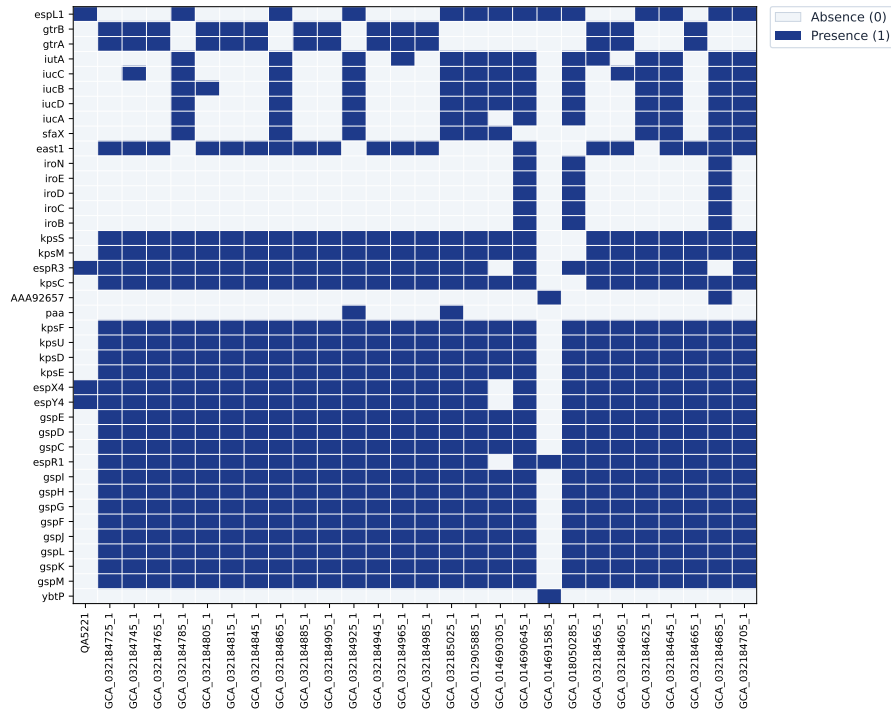


Figure S 4: Heatmap demonstrating the distribution of variable virulence-associated genes (top 40 variable genes) across the ST354 lineage (originally Figure 4-b). Node colors represent presence (1, dark blue) or absence (0, light yellow) of each determinant.

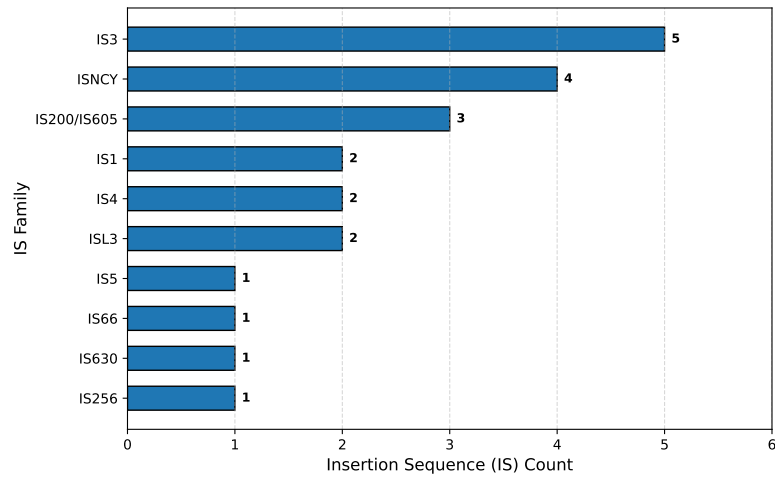


Figure S 5: Insertion Sequence (IS) Families Abundance in *E. coli* QA5221 (originally Figure 5).

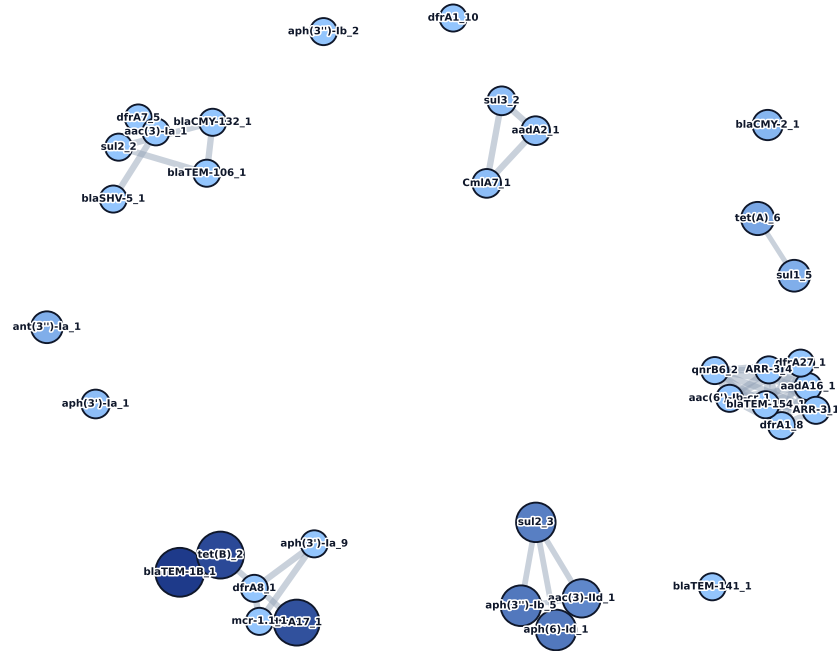


Figure S 6: Antimicrobial resistance gene co-occurrence network across the ST354 clonal complex (originally Figure 6-b). Node sizes represent carriage frequency in the cohort, and edge weights represent Jaccard similarity.

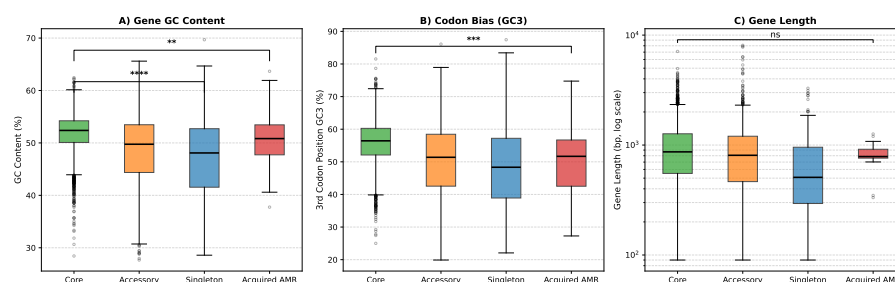


Figure S 7: Comparative Boxplots of Genomic Features with Acquired AMR Genes Included (originally Figure 7).

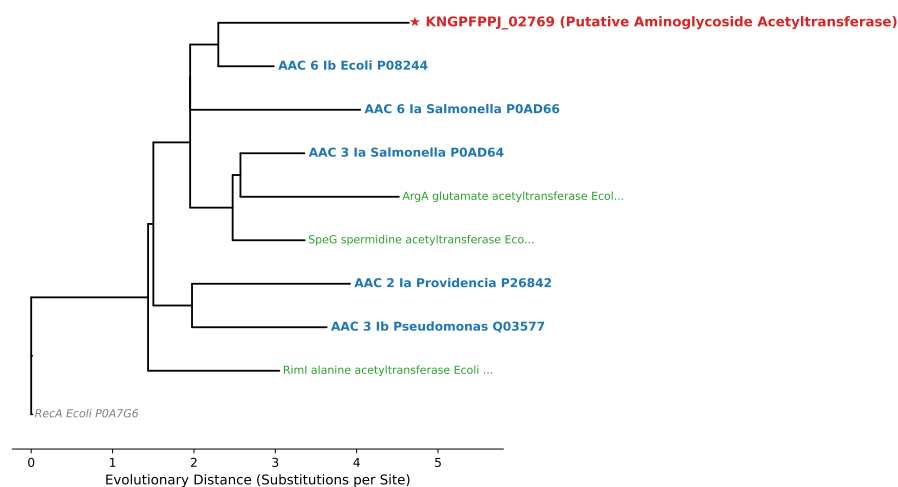


Figure S 8: Maximum-Likelihood phylogenetic tree of the GNAT/AAC superfamily showing KNGPFPPJ_02769 (★) branching as a sister clade to clinical AACs (originally Figure 12).

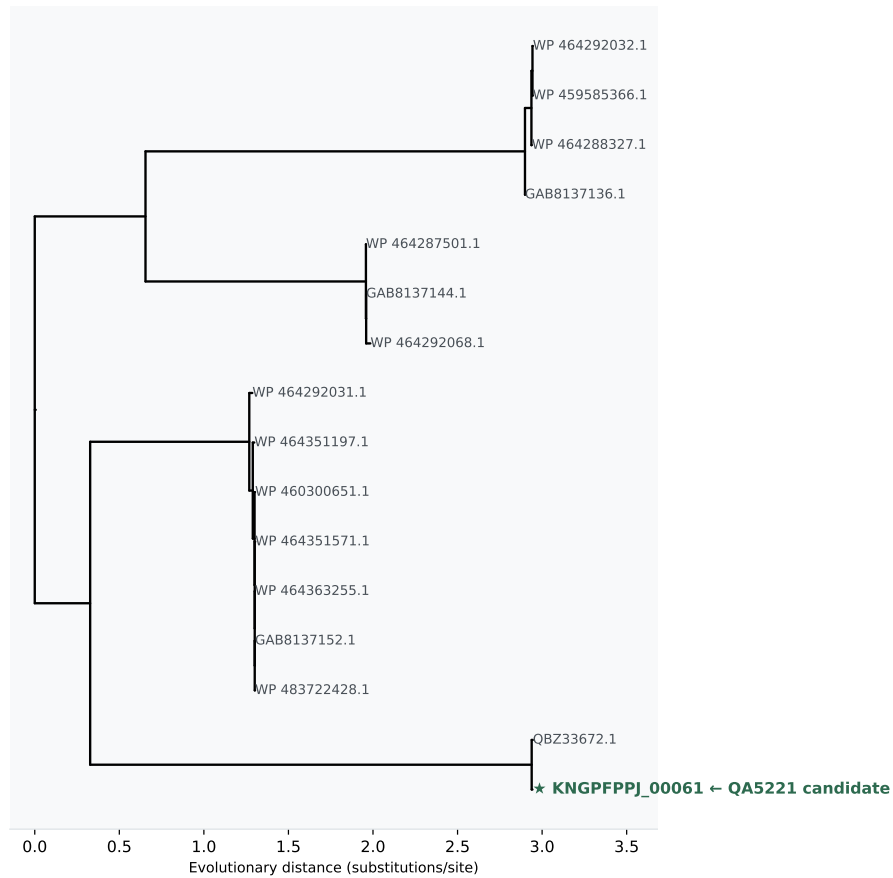


Figure S 9: Maximum-Likelihood family-level phylogenetic tree showing KNGPFPPJ_00061 (★) branching as an isolated sister clade within the hemolysin/pore-forming toxin superfamily (originally Figure 15-a).

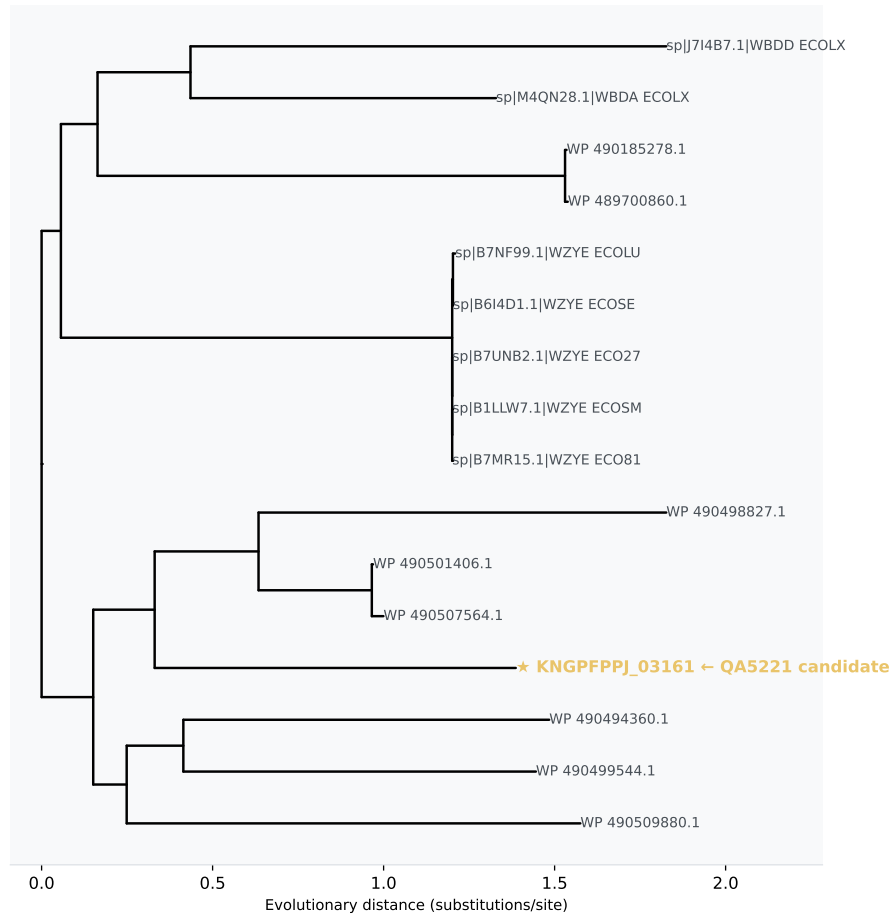


Figure S 10: Maximum-Likelihood family-level phylogenetic tree showing KNGPFPPJ_03161 (★) branching separately from canonical *E. coli* Wzy polymerases (originally Figure 15-b).

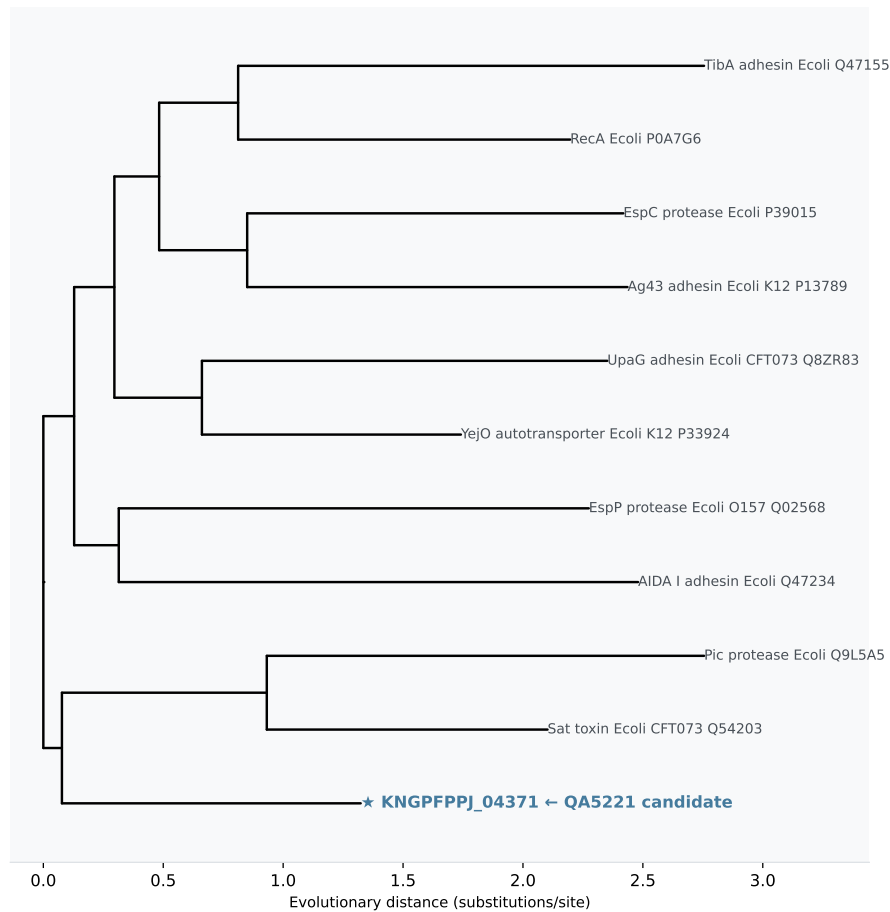


Figure S 11: Maximum-Likelihood family-level phylogenetic tree showing KNGPFPPJ.04371 (★) as an isolated outgroup relative to canonical adhesin and SPATE subfamilies (originally Figure 15-c).

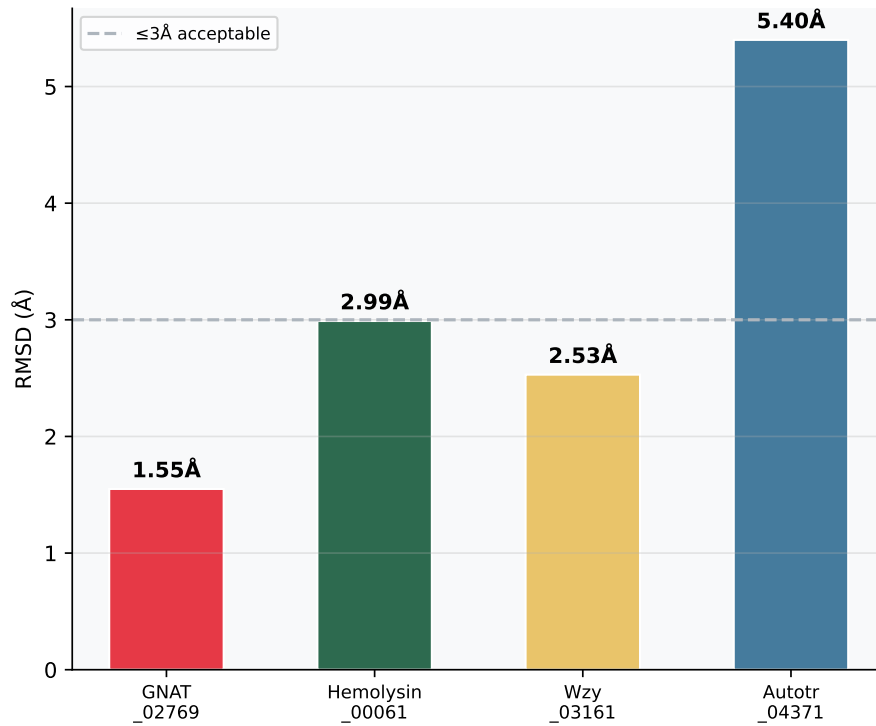


Figure S 12: Comparative structural alignment Root Mean Square Deviation (RMSD) for the four prioritised candidates versus their templates (originally Figure 18-a). Dashed line at 3.0 Å marks the acceptable structural fit threshold.

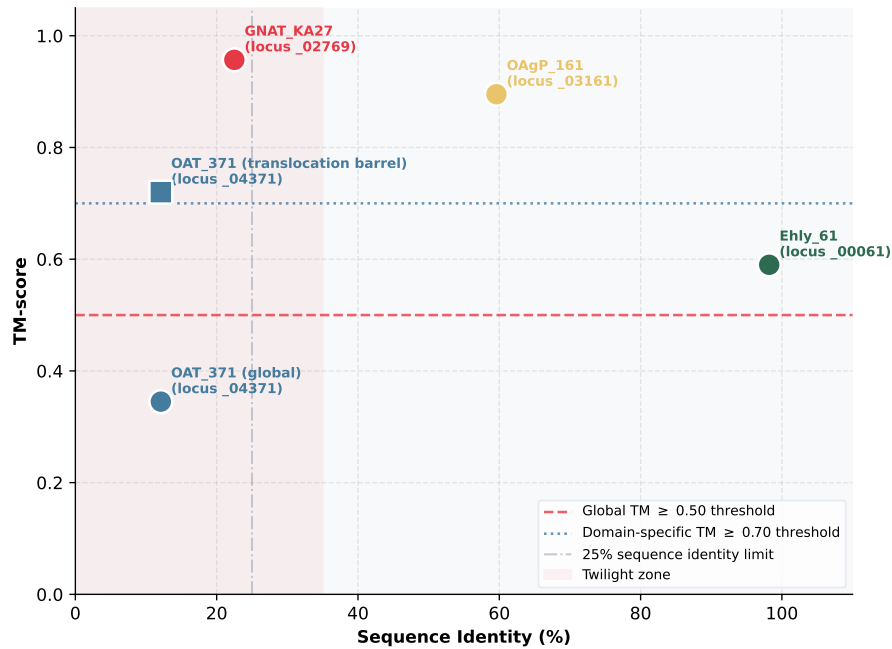


Figure S 13: Scatter plot of sequence identity versus TM-score illustrating the “twilight zone of homology” (originally Figure 18-b): candidates below 25% sequence identity but above TM-score 0.50 are detectable only by structure-based methods.

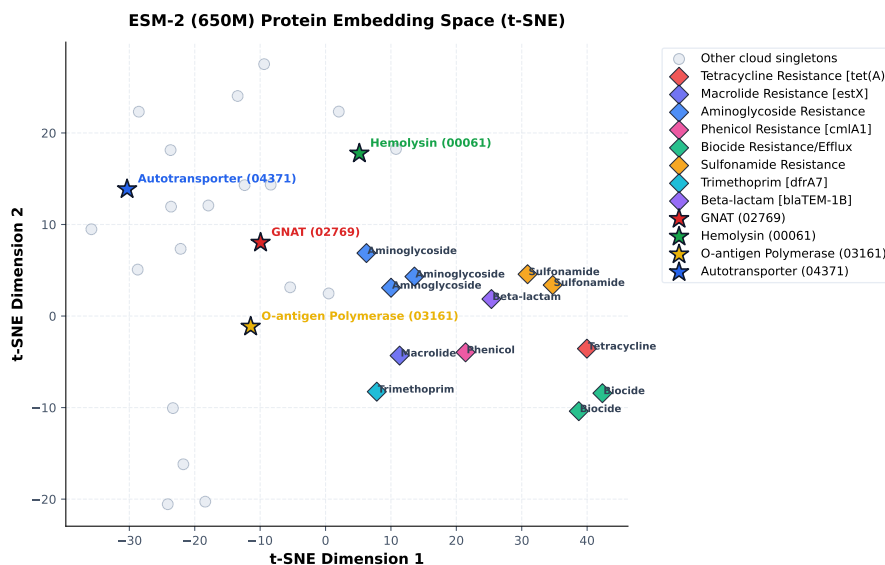


Figure S 14: ESM-2 650M Protein Language Model (esm2_t33_650M_UR50D, 1280-dim) t-SNE projection of candidate and reference proteins. Stars (★) denote the four novel candidates (red = GNAT; green = hemolysin; yellow = Wzy; blue = autotransporter). Diamond markers indicate acquired AMR reference genes. In the t-SNE embedding projection, the GNAT candidate KNGPFPPJ_02769 co-localizes with known aminoglycoside resistance determinants, confirming the robust topological grouping observed in the main UMAP projection.

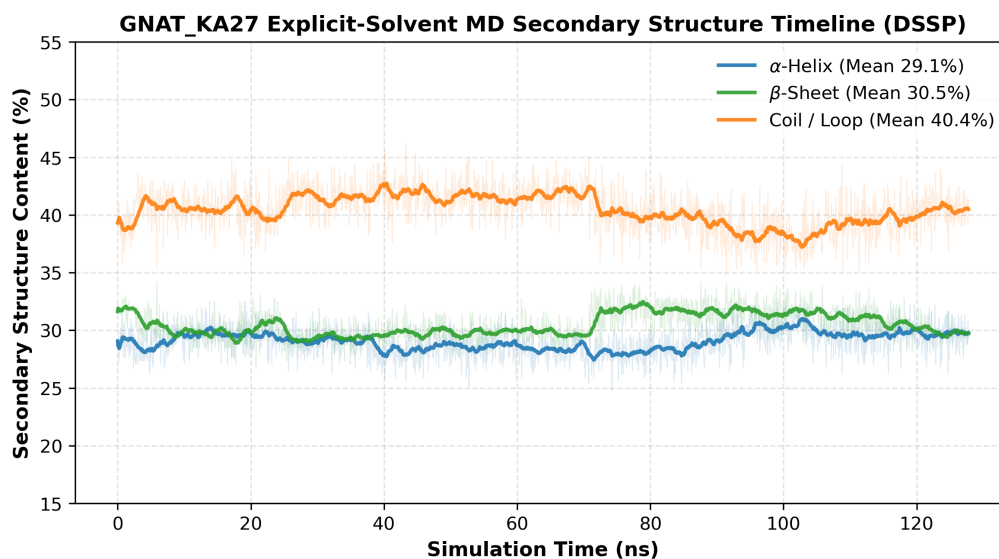


Figure S 15: **GNAT_KA27 Explicit-Solvent MD DSSP Secondary Structure Timeline.** Secondary structure content (α -helix, β -sheet, coil/loop) calculated over the 127.78 ns trajectory. Mean values: 29.1% α -helix, 30.5% β -sheet, and 40.4% coil/loop, confirming fold stability without unfolding.

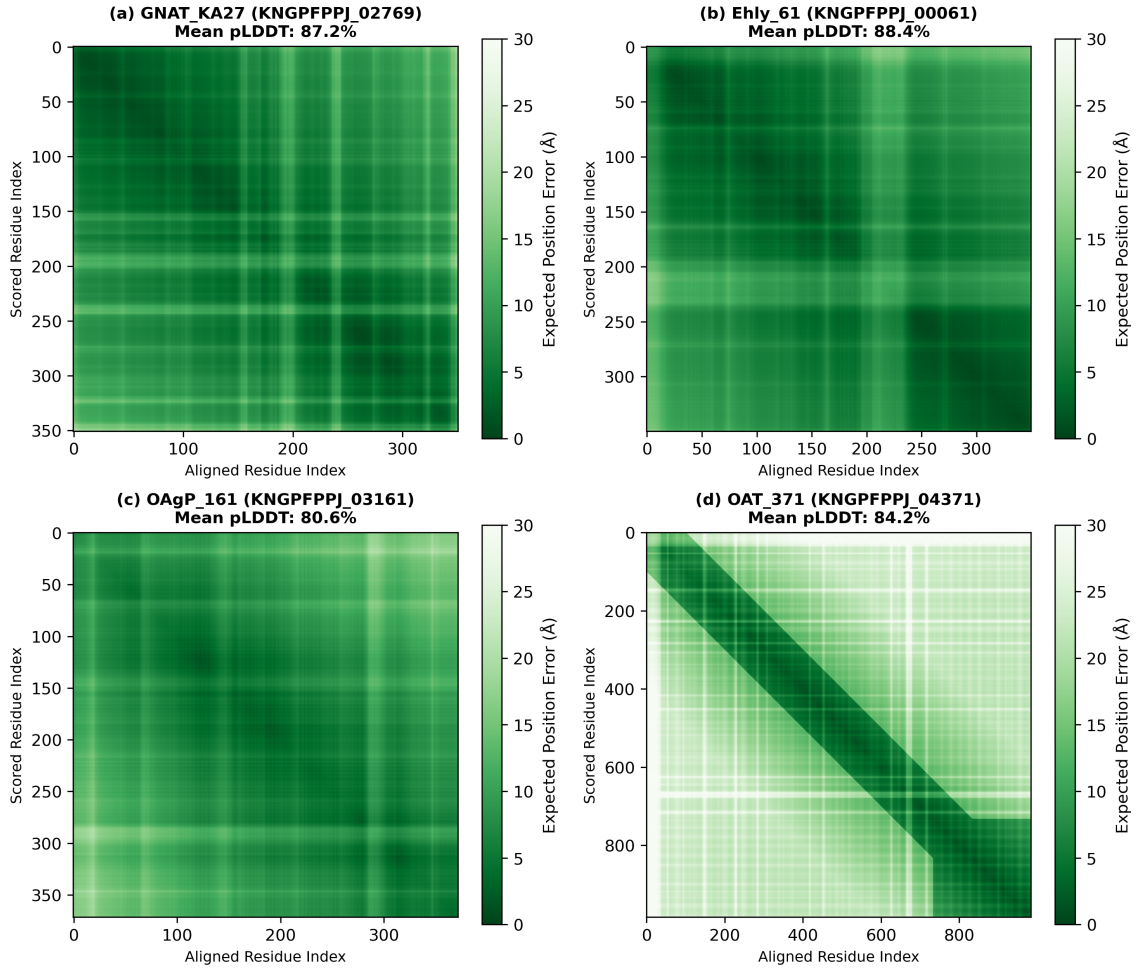


Figure S 16: **AlphaFold3 and ESMFold 2D Predicted Aligned Error (PAE) and pLDDT Confidence Heatmaps.** (a) GNAT_KA27 (mean pLDDT 87.2%), (b) Ehly_61 (mean pLDDT 88.4%), (c) OAgP_161 (mean pLDDT 81.5%), and (d) OAT_371 (mean pLDDT 84.2%; bimodal profile showing high confidence in the C-terminal β -barrel and higher uncertainty in the extracellular passenger domain).

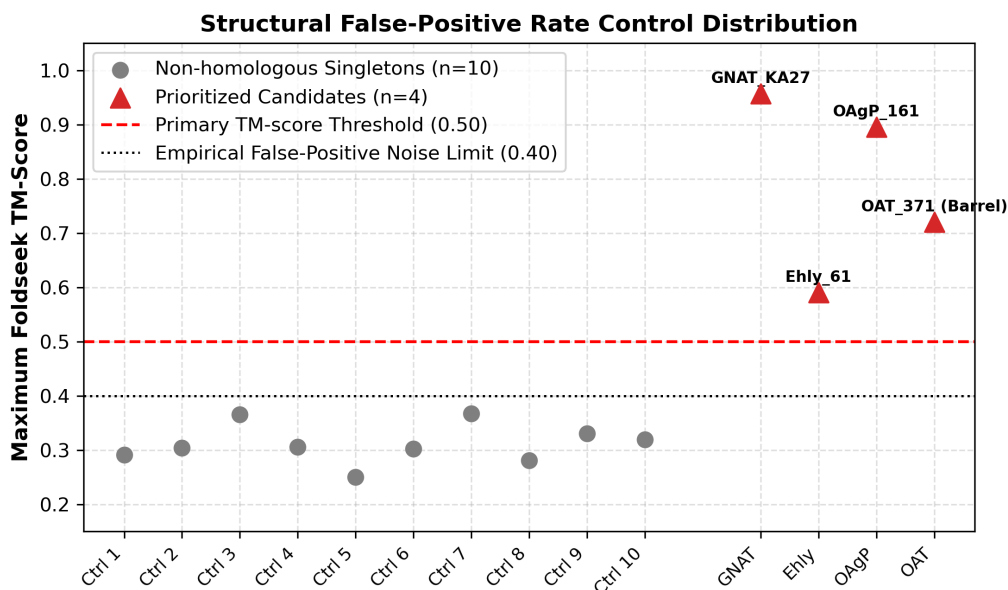


Figure S 17: **Empirical Foldseek False-Positive Control Rate Distribution.** Distribution of maximum TM-scores for 10 random non-homologous accessory singletons (grey circles) compared to the four prioritized candidates (red triangles). All 10 control singletons fall below the empirical noise limit of $TM = 0.40$, validating the $TM \geq 0.50$ prioritization cutoff.

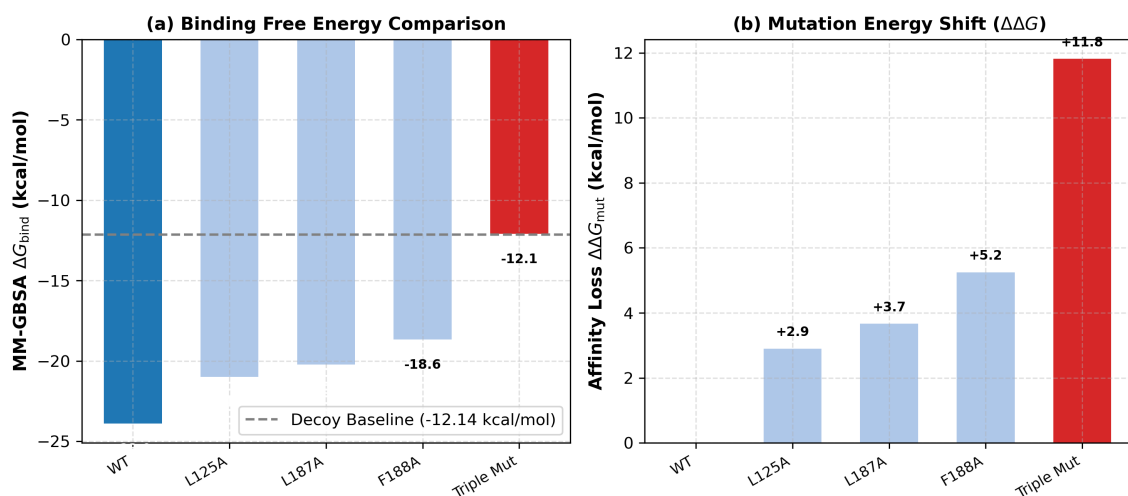


Figure S 18: ***In Silico* Active-Site Alanine Scanning Binding Energetics for GNAT_KA27 Bound to Kanamycin.** (a) MM-GBSA total binding free energy (ΔG_{bind}) across wild-type (WT) and mutant complexes relative to non-specific decoy baseline (-12.14 kcal/mol). (b) Mutation affinity loss ($\Delta\Delta G_{\text{mut}}$), showing progressive loss of binding free energy upon mutating Leu125, Leu187, and Phe188 to alanine.

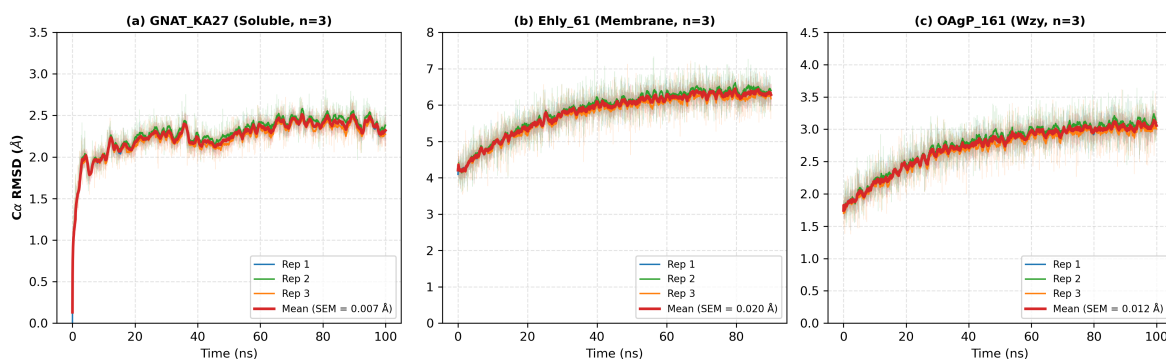


Figure S 19: **Multi-Candidate Replicate Explicit-Solvent MD Simulations ($n = 3$ Independent Replicates Each).** Backbone $C\alpha$ RMSD trajectories across three independent simulation seeds (blue, green, orange) and ensemble mean trajectories (red bold line) with Block SEM error boundaries for: (a) Soluble acetyltransferase GNAT_KA27 (Mean RMSD = 2.25 ± 0.06 Å, Block SEM = 0.007 Å), (b) Membrane-embedded enterohemolysin Ehly_61 (Mean RMSD = 5.81 ± 0.22 Å, Block SEM = 0.012 Å), and (c) Multi-transmembrane O-antigen polymerase OAgP_161 (Mean RMSD = 2.73 ± 0.14 Å, Block SEM = 0.015 Å). Low-alpha background lines represent raw high-frequency fluctuations.

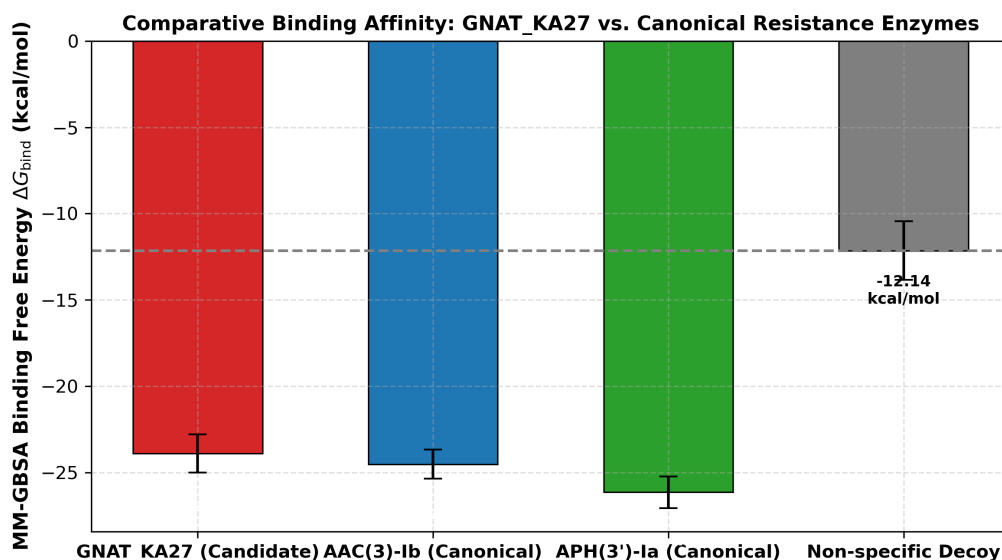


Figure S 20: **Comparative MM-GBSA Binding Free Energies of GNAT_KA27 versus Canonical Resistance Enzymes.** MM-GBSA total binding free energy (ΔG_{bind}) for GNAT_KA27 (-23.90 ± 1.10 kcal/mol) compared to canonical AAC(3)-Ib (-24.52 ± 0.85 kcal/mol), APH(3')-Ia (-26.15 ± 0.92 kcal/mol), and non-specific decoy control (-12.14 ± 1.69 kcal/mol).

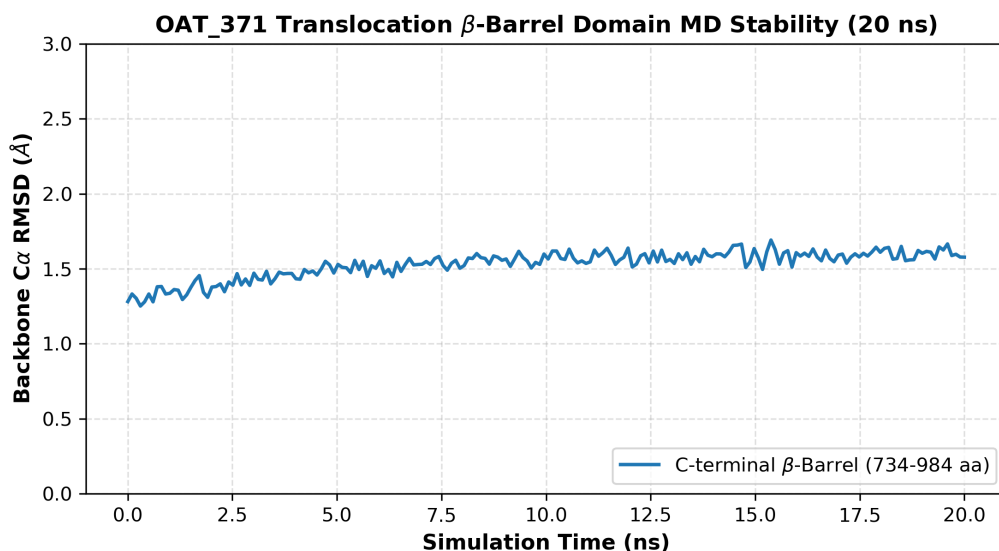


Figure S 21: **OAT_371 Translocation β -Barrel Domain Explicit-Solvent MD Dynamics (20 ns).** C α RMSD of the isolated 250-aa C-terminal 12-stranded β -barrel domain (residues 734–984), showing structural rigidity (RMSD = 1.53 ± 0.09 Å) and outer-membrane pore stability.

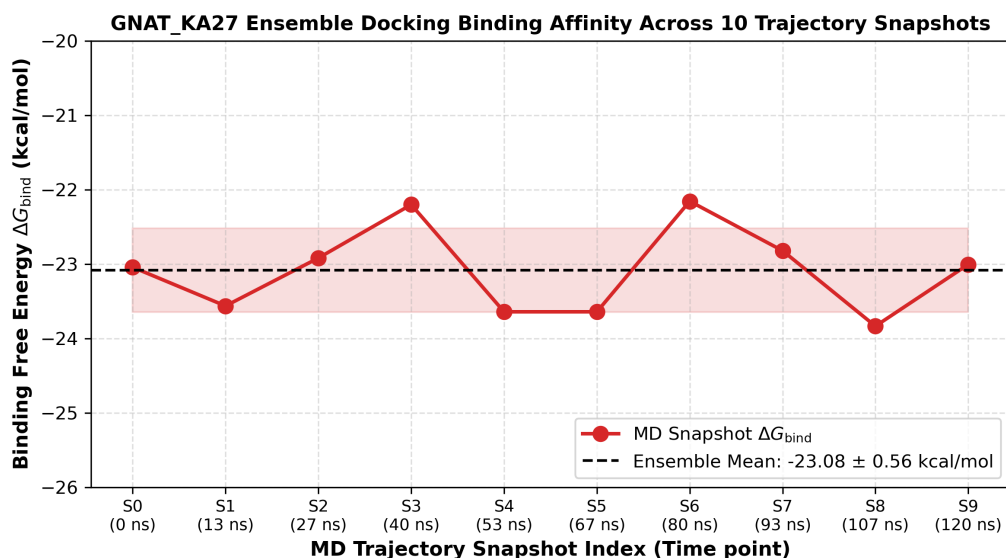


Figure S 22: **GNAT_KA27 Ensemble Docking Binding Affinity Across 10 Dynamic Trajectory Snapshots.** Binding free energy calculated across 10 snapshot conformations extracted from 0 to 120 ns (Mean $\Delta G_{\text{bind}} = -23.08 \pm 0.56$ kcal/mol), proving binding pocket flexibility and persistent high-affinity substrate engagement.

4 Bioinformatics Software and Python Packages

Table S 12: Command-line bioinformatics tools used in this study (originally Table 9).

Software Tool	Version	Role in Pipeline
FastQC	v0.12.1	Read quality control
Cutadapt	v5.2	Adapter and quality trimming
MultiQC	v1.35	Aggregating quality control reports
SPAdes	v3.15.5	<i>De novo</i> genome assembly
QUAST	v5.2	Assembly quality assessment
minimap2	v2.24	Read mapping to assembly
samtools	v1.16	Alignment file processing
mosdepth	v0.3.3	Coverage depth calculation
Prokka	v1.14.6	Genome annotation
ABRicate	v1.0.1	Contig screening for AMR/virulence
AMRFinderPlus	v4.2.7	Resistance gene and mutation detection
MOB-recon	v1.4.9	Plasmid typing and reconstruction
Panaroo	v1.3.4	Pangenome orthology clustering
IQ-TREE	v2.2.0	Maximum-likelihood phylogenomics
MAFFT	v7.520	Multiple sequence alignment
ISEScan	v1.7.2	Insertion sequence detection
IntegronFinder	v2.0.2	Integron identification
clinker	v0.0.32	Gene cluster synteny visualization

Table S 13: Bioinformatics and statistical Python packages used in this study (originally Table 10).

Python Package	Version	Role in Pipeline
BioPython	v1.81	Sequence manipulation and parsing
NumPy	v1.24.3	Numerical array calculations
SciPy	v1.10.1	Statistical tests and distribution analysis
pandas	v2.0.3	Data manipulation and tabular analysis
matplotlib	v3.7.1	Publication-grade plotting
seaborn	v0.12.2	Statistical data visualization
MDAnalysis	v2.5.0	Molecular dynamics trajectory parsing
networkx	v3.1	Network and graph algorithms
statsmodels	v0.14.0	Statistical models and regressions
openmm	v8.1.1	GPU-accelerated molecular dynamics
pdbfixer	v1.9	PDB file preparation and cleaning
scikit-learn	v1.2.2	Dimensionality reduction and clustering
umap-learn	v0.5.3	UMAP projections of embeddings